Ailing Jin

Department of Geography, Sustainability, Community, and Urban Studies
University of Connecticut
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EDUCATION

2021-Dec. 2026 (expected)	**PhD in Geography** University of Connecticut (UConn), Storrs, USA *Dissertation: "Fine-Scale Assessment of Personal Exposure to Indoor and Outdoor Air Pollution Using Geospatial and Real Time Sensing Technologies"*
2021-2023	**MS in Health Promotion Sciences** UConn, Storrs, USA
2018-2021	**MA in Geography** Binghamton University, State University of New York (SUNY), Binghamton, USA
2014-2018	**BS in Ecology** Minzu University of China, Beijing, China

POSITIONS & EMPLOYMENT

2025	**Instructor**, Institute for Student Success, UConn
2022-present	**Teaching Assistant**, Department of Geography, Sustainability, Community, and Urban Studies (GSCU), UConn
2021-2023	**Research Assistant**, Department of Allied Health Sciences, UConn

RESEARCH & TEACHING INTERESTS

Geographic Information Science (GIS)	Indoor/Outdoor Air Pollution
Spatiotemporal Data Analysis	Exposure and Health Risk Assessment
Mobile Sensing Technologies	Human Mobility

PUBLICATIONS

Jin, A., Park, Y. M., Chen, X., Liang, L., & Gong, P. (2026). Leveraging real-time personal monitoring of air pollution and physical activity levels, and location tracking for mobility-and inhaled dose-based assessments of exposure and health risk. *Environmental monitoring and assessment*, *198*(3), 242.
Link: *https://link.springer.com/article/10.1007/s10661-026-15090-x*

- *Supported by the **Summer Research Fellowship** from the Department of GSCU at UConn.*
- *Selected as a **Finalist** for the Health and Medical Geography Specialty Group (HMGSG) Peter Gould Student Paper Award at the 2025 American Association of Geographers Annual Meeting (AAG)*

Jin, A., Chen, X., Huang, X., Li, Z., Caspi, C. E., & Xu, R. (2023). Selective Daily Mobility Bias in the Community Food Environment: Case Study of Greater Hartford, Connecticut. *Nutrients*, 15(2), 404.
Link: *https://www.mdpi.com/2072-6643/15/2/404*

- *Won **First Place** in the Lightning Talk Competition at the University Consortium for Geographic Information Science (UCGIS)*

Chen, X., **Jin, A.**, & Xu, R. (2023). Examining Human Mobility Patterns in Restaurant Visits: A Social Vulnerability Perspective. *Proceedings of the Northeast Travel and Tourism Research Association (NETTRA) Annual Conference 2023. Philadelphia, Pennsylvania, April 14–16*
Link*:https://www.nettra.org/uploads/1/3/4/1/134158139/nettra_2023_annual_conference_proceedings__chen.pdf*

PAPERS IN PROGRESS

Jin, A., Park, Y. M., & Chen, X. Mobility-Based Indoor $PM_{2.5}$ Exposure Assessment Using an Indoor Air Quality Sensor Network and Bluetooth Low Energy (BLE)-Based Localization. Target submission date: (Jan. 2027)

- *Recipient of the AAG International Geographic Information Fund (IGIF) **Graduate Research Award***

Jin, A., Park, Y. M., Moore, T. E., Bae, E., & Chen, X. Modeling Spatiotemporal Variations of Indoor $PM_{2.5}$ Using High-Resolution Indoor Sensor Network Measurements. Target submission date: (Nov. 2026)

RESEARCH EXPERIENCE

2025-present — Department of GSCU, UConn
Topic: *Fine-Scale Spatiotemporal Modeling of Indoor Air Pollution and Personal Exposure: Integrating Indoor Mobility Tracking and Sensor Networks.*

- Conduct spatiotemporal modeling of indoor air pollution across an apartment unit and create indoor air quality maps.
- Develop a data collection protocol for high-resolution, individual-level geospatial data.
- Recruit participants, obtain informed consent, and administer surveys.
- Establish indoor air sensor networks across participants' homes in conjunction with the indoor positioning systems (i.e., Bluetooth Low Energy beacon systems) for real-time monitoring of indoor mobility and $PM_{2.5}$ concentrations.
- Generate participant reports on exposure using GIS and share the reports with participants.
- Analyze collected datasets using GIS and lead manuscript writing.

2023-2025 — Department of GSCU, UConn
Topic: *Mobility- and Inhaled Dose-Based $PM_{2.5}$ Exposure and Health Risk Assessment.*

- Improved personal air pollution exposure and health risk assessment by integrating real-time indoor and outdoor $PM_{2.5}$ monitoring data with individuals' real-time physical activity intensity, Global Position System (GPS) tracking, and physiological

characteristics.
- Conducted statistical analyses to compare different exposure metrics across different population groups.
- Developed 3D geovisualizations to depict the spatiotemporal dynamics of exposure.
- Submitted research findings to a peer-reviewed journal as the first author.

2021-2023 Department of Allied Health Sciences, UConn
Topic: *Community Food Environment Assessment Using High-Resolution Location Data from Mobile Phones.*
- Conducted multivariate linear regressions to examine the relationship between community-level socioeconomic factors and restaurant patronization, and revealed the existence of a behavior-related uncertainty— selective daily mobility bias—within community-based food environments.
- Created and designed a map of restaurant visit patterns using GIS.
- Published research findings in the peer-reviewed journal as the first author.

2019-2021 Department of Geography, Binghamton University, SUNY
Topic: *Storefront Transparency Evaluation in Commercial Areas Using Google Street View (GSV) and Deep Learning.*
- Extracted window and wall information from GSV images using deep learning to evaluate building transparency along Main Street.

TEACHING & MENTORING EXPERIENCE

Department of GSCU, UConn
Jan.2026-Current
Research Team Mentor: Led and mentored a team of six undergraduate research assistants in a household-based study of indoor air pollution and personal exposure. Trained the team in the study protocol, indoor sensor network installation, standardized data collection, and participant communication to support consistent, high-quality fieldwork.

Institute for Student Success, UConn
Sep. 2025-Dec. 2025
Instructor of Record: First Year Experience University Learning Skills (Fall 25)
- Independently designed and delivered a first-year university learning skills course.
- Mentored undergraduate students on academic strategies and time management.
- Mentored undergraduate teaching assistants and coordinated course activities.

Department of GSCU, UConn
Jan. 2025–May 2025
Undergraduate Research Mentor: Supervised an undergraduate student in project design, GPS tracking, and low-cost air sensor data collection, data analysis, result interpretation, and a poster presentation at an academic conference.

Department of GSCU, UConn
Aug. 2022-present
Teaching Assistant: Global Urbanization (Fall 22; Spring 25) (Online/In person) | World Regional Geography (Fall 23; Spring 24; Fall 24; Fall 25) (Online/In person) | Globalization (Spring 23; Fall 23; Fall 25) (Online/In person)
- Graded quizzes, exams, and assignments.

- Held office hours, meeting with students to address course-related questions and concerns.
- Assisted in organizing in-class discussions and provided support during in-class assignments and exams.

Department of GSCU, UConn
Mar. 2023 & Feb. 2025
Guest Lecturer: Applications of GIS (Spring 23; Spring 25)
- Designed and delivered lectures for undergraduate students on using GIS in environmental exposure assessments.
- Engaged students in discussions and facilitated Q&A sessions, addressing questions and clarifying concepts.

AWARDS & SCHOLARSHIPS

2026 **Summer Doctoral Dissertation Fellowship,** UConn ($2,000)
2026 **Summer Research Fellowship,** Department of GSCU, UConn ($900)
2026 **Graduate Research Award,** IGIF, AAG ($600)
2025 **Finalist**, HMGSG Peter Gould Student Paper Award Competition, AAG
2025 **Summer Research Fellowship**, Department of GSCU, UConn ($1800)
2025 **Urban Scholars Fellowship**, Department of GSCU, UConn ($500)
2024 **Summer Research Fellowship**, Department of GSCU, UConn ($1800)
2024 **Conference Participation Award**, UConn ($750)
2023 **Summer Research Fellowship**, Department of GSCU, UConn ($1800)
2023 **First Place**, Lightning Talk Competition, UCGIS ($300)
2022 **Summer Research Fellowship**, Department of GSCU, UConn ($1800)
2021 **Outstanding Presentation Award**, Connecticut GIS Day ($100)
2018 **Undergraduate Research Innovation Award**, Minzu University of China

PRESENTATIONS

Feb. 2027	Modeling Spatiotemporal Variations of Indoor $PM_{2.5}$ Using High-Resolution Indoor Sensor Network, AAG Annual Meeting 2026, New York City, New York, USA
Mar. 2026	"Mobility-Based Indoor $PM_{2.5}$ Exposure Assessment Using an Indoor Air Quality Sensor Network and Bluetooth Low Energy (BLE)-Based Localization", AAG Annual Meeting 2026, San Fransico, California, USA
Mar. 2025	"Leveraging Real-Time Monitoring of Air Pollution, Physical Activity Levels, and Geospatial Positioning for Human Mobility- and Inhaled Dose-Based Health Risk Assessments", AAG Annual Meeting 2025, Detroit, Michigan, USA
Sep. 2024	"From Concentration to Inhalation: Examining Group Differences in Health Risks of $PM_{2.5}$ Exposure Using Portable Sensing Data", New England – St. Lawrence Valley Geographical Society Fall 2024 Conference, Westfield, MA, USA
Jun. 2023	"Selective Daily Mobility Bias in the Community Food Environment: Case Study of Greater Hartford, Connecticut", UCGIS 2023 Symposium, New Heaven, Connecticut, USA
Mar. 2023	"Selective Daily Mobility Bias in the Community Food Environment: Case Study of Greater Hartford, Connecticut", AAG Annual Meeting 2023, Denver, Colorado, USA
Nov. 2021	"Storefront Transparency Evaluation in Commercial Areas Using Google Street View and Deep Learning", Connecticut GIS Day, Storrs, CT, USA

PROFESSIONAL SERVICE

Manuscript Reviewer

- Computational Urban Science
- npj Digital Medicine